Utpal Kant Sharma

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PROFESSIONAL SUMMARY

Computer Science graduate building LLM-backed systems with Python, FastAPI, and LangGraph. Developed a multi-provider LLM gateway with failover, semantic caching, and evaluation; human-in-the-loop agent workflows; and local multimodal RAG with measured retrieval improvements.

EXPERIENCE

Assam Electronics Development Corporation Ltd (Amtron) **Assam, India**
Data Science Intern *Dec 2024 – Jan 2025*

- Optimized complex SQL queries involving JOIN operations, aggregations, and composite indexing across 10,000+ record relational databases, reducing query execution time from 45 seconds to under 15 seconds.
- Developed ML pipelines using Python, Pandas, and NumPy to process structured datasets and extract operational insights.
- Built Matplotlib dashboards to visualize quarterly operational KPIs and translate SQL outputs into internal reports.

PROJECTS

TraceGate – LLM Gateway [GitHub] *FastAPI, SQLAlchemy, SQLite, Ollama, Groq, Gemini*

- Architected a self-hosted, OpenAI-compatible LLM gateway using FastAPI, integrating Ollama, Groq, and Gemini with priority-based automatic failover.
- Implemented embedding-based semantic caching using Sentence-Transformers and cosine similarity, achieving ∼60× cache-hit latency reduction (665ms to 11ms).
- Engineered bearer-token authentication, per-key token-budget enforcement, and asynchronous SQLite trace logging for latency, token usage, and provider monitoring.
- Built a CLI-based LLM evaluation harness using exact-match, semantic similarity, and LLM-as-judge to benchmark models and identify scoring discrepancies.

NeuroLeads AI – B2B Sales Orchestrator [GitHub] *LangGraph, LangChain, PostgreSQL, Docker*

- Built a multi-agent LLM pipeline using LangGraph with conditional routing for research, structured extraction, drafting, and a self-critique loop.
- Engineered defensive parsing to handle malformed non-JSON LLM responses and configured reasoning-effort parameters to optimize API token consumption.
- Containerized a FastAPI backend with Postgres-backed LangGraph checkpointing to enable a persistent, human-in-the-loop approval workflow across separate API requests.

Multimodal RAG over PDFs with Local LLMs [GitHub] *Docling, Ollama, Qdrant, FastAPI, Streamlit*

- Built a fully local multimodal RAG pipeline using Docling, Ollama, and Qdrant to answer cited questions over PDFs containing text, tables, charts, and image-only slides.
- Designed chunk-level retrieval evaluation on a 27-question golden set, exposing missing chart evidence despite 100% page-level recall; improved table chunk ranks from 17/11/18 to 5/2/9 with structured row embeddings.
- Recovered chart values via PDF text-layer extraction (0/6 to 6/6 indexed) and filtered decorative images, reducing index size from 177 to 76 chunks.

TECHNICAL SKILLS

Languages: Python, SQL, C.
AI & LLM Frameworks: LangChain, LangGraph, Agentic AI, RAG, Prompt Engineering, Fine-tuning, MCP.
Models & Vector DBs: OpenAI, Gemini, Claude, Groq, Hugging Face, ChromaDB, Qdrant.
Backend & Cloud: Django, DRF, FastAPI, REST APIs, PostgreSQL, Git, GitHub, Docker, AWS (EC2, S3), Vercel.

EDUCATION

The Assam Kaziranga University **Guwahati, Assam**
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.08/10 *Aug 2022 – May 2026*

- **Coursework:** Data Structures & Algorithms, Machine Learning, Database Management Systems, Operating Systems, Computer Networks.

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate (Coursera Credential ID: YB15NDMCH302) – Covered RAG architectures, agentic workflows, CrewAI, AutoGen, and MCP.